



FOR YOUR DEVELOPER

# Every piece of technology the company needs

This document is for your developer.

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Status labels apply throughout: KNOWN, VERIFIED, ASSUMPTION, ESTIMATE, TO VALIDATE, PROPOSED. Nothing in this document claims traction, approvals, partners or revenue that do not exist. Governing file: FACTS\_BASE.

**PREPARED FOR**

The Founder, Vytalix

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**This document is for your developer.**

**One table for the whole company: what each thing is for, what to use, roughly what it costs a month, and what else would do.**

**In one minute**

- Vytalix has two separate estates: cheap bought tools for the business, and one built platform for MaternaLink.
- Buy everything that is not the product. Build only the product.
- The business tools cost £90–£250 a month at the start, rising to £460–£1,560 once the build is running.
- MaternaLink runs on AWS in London for £350–£700 a month at first release.
- The "do not build yet" list at the end matters as much as the buy list. Keep it on the agenda.

**What to do**

| Action                                                                                                                                 | Who       | By when                                |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------|
| On incorporation: domain, Google Workspace, email security, password manager, HubSpot free, Xero, Webflow, Plausible, ICO registration | Founder   | One week of setup, under £300 a month  |
| Set up the AWS organisation, infrastructure-as-code baseline and GitHub before the build starts                                        | Developer | Before build week 1                    |
| Write down the AI policy and have the founder sign it                                                                                  | Founder   | Before any tool is used on client work |
| Choose the e-learning platform when the first course is scheduled, not before                                                          | Founder   | Month 6                                |
| Book Cyber Essentials for the quarter in which the MaternaLink test environment exists                                                 | Developer | 2027                                   |

**How it fits together**

## flowchart TB

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subgraph Bought[Bought: business tools, never any patient data]
  WEB[Website: Webflow now, Next.js later] --> CRM[HubSpot]
  LMS[Courses: Thinkific or LearnWorlds] --> FIN[Xero, Stripe, GoCardless, Paystack]
  GW[Google Workspace: mail, docs, sign-in] --> CRM
  GW --> SEC[1Password, device management, contracts]
  WEB --> AN[Plausible analytics]
end
subgraph Built[Built: MaternaLink on AWS London, all patient data]
  PWA[Woman and staff web apps] --> API[API and background workers]
  API --> PG[(PostgreSQL)]
  API --> S3[(File storage)]
  API --> KC[Keycloak sign-in]
  API --> CH[SMS and WhatsApp adapters]
  PG --> MB[Metabase dashboards]
end
API --> LATER[Deferred: NHS interfaces, DHIS2 export, Nigeria hosting, data warehouse, customer portal]

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*Caption: business tools on the left are bought and hold no patient data; the product platform on the right is built, and patient data never leaves it.*

## The whole company's technology in one table

All prices are ESTIMATE (our best guess, not a fact) in pounds at September 2026, from public list prices. Check every one before buying; vendors rename plans often.

| What it is for     | What to use                                                                                                                                                                     | Roughly per month                           | The alternative                                                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company website    | Webflow with its CMS plan in year 1. Next.js with a headless CMS once engineers exist in 2027                                                                                   | £18–£25, plus £10–£15 a year for the domain | Framer, similar and design-led. WordPress, cheap but a security burden. Squarespace, too limited                                                                                            |
| Marketing content  | Webflow's own CMS now, then Sanity or Payload later. MaternaLink's clinical content is not here: it lives in the product with its own two-person approval                       | Included, then £0–£80                       | Contentful, pricier. Strapi, self-hosted                                                                                                                                                    |
| Sales and contacts | HubSpot, free tier then Starter. The investor list imports as its own pipeline                                                                                                  | £0, then £15–£20 per seat                   | Pipedrive £12–£14 per seat, sales only. Attio £25+. Salesforce, no                                                                                                                          |
| Customer portal    | Do not build in year 1. Serve consulting clients with shared drives and PandaDoc, learners inside the course platform, MaternaLink customers inside the product's admin screens | £0 now, £30k–£60k to build later            | HubSpot's own portal for tickets                                                                                                                                                            |
| Selling courses    | Thinkific if the first courses are for individuals, LearnWorlds if a corporate or NHS group licence is signed first                                                             | £30–£120 Thinkific, £25–£250 LearnWorlds    | Teachable, weaker on corporate. Moodle, only when a university or NHS buyer demands it: £80–£150 hosted, or £30–£100 self-hosted plus admin time. Custom build £40k–£80k, not before year 3 |

| What it is for               | What to use                                                                                                                                                                                                                      | Roughly per month                                                                                                          | The alternative                                                                                                                                               |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The MaternaLink product      | TypeScript throughout; Next.js web apps; a NestJS API; PostgreSQL with row-level security; Redis for jobs; S3; Keycloak; AWS eu-west-2 (London); Terraform; GitHub Actions; SMS adapters for Africa's Talking, Termii and Twilio | £350–£700 at first release; £1,000–£2,500 at 5 organisations and 20,000 women                                              | Azure UK South, equivalent and worth choosing if the first NHS customer is Microsoft-centred. Do not run both. GCP London has fewer NHS references            |
| Product admin and dashboards | Admin lives inside the MaternaLink staff app. Business dashboards on Metabase, self-hosted, reading a copy of the database. Company numbers start as a spreadsheet fed by HubSpot, Xero and course exports                       | £30–£50 for Metabase, £0 for the spreadsheet                                                                               | Retool, fast but data leaves the region unless self-hosted. Power BI if the buyers are Microsoft-heavy                                                        |
| Analytics                    | Plausible for the website, hosted in the EU with no cookies. For the product, an events table inside our own database holding no personal data, read by Metabase                                                                 | £9–£15, product side included in cloud cost                                                                                | Fathom. Google Analytics is free but brings a consent banner and data-transfer questions. Amplitude and Mixpanel are excluded: data leaves the region         |
| Staff sign-in                | Google Workspace as the identity provider for every business tool, with security keys for admins. A company domain replaces the personal Gmail on day one                                                                        | £5–£11 per user                                                                                                            | Microsoft 365 is a legitimate choice because NHS buyers are Microsoft-heavy. Decide once                                                                      |
| MaternaLink sign-in          | Keycloak, self-hosted in London, for staff. Phone code plus device PIN for women                                                                                                                                                 | £40–£70                                                                                                                    | Auth0 and Clerk are easier but their UK data residency is unconfirmed. Cognito is cheap but clumsy to federate                                                |
| Taking payments              | Stripe for cards and subscriptions, usually through the course platform. Xero invoicing with GoCardless direct debit for retainers. Paystack or Flutterwave for Nigeria, which needs a Nigerian entity or a partner              | Xero £15–£40. Fees: about 1.5% plus 20p on UK cards; GoCardless about 1% capped; Paystack about 1.5% plus ₦100 over ₦2,500 | Paddle handles global VAT itself at a higher percentage. Interswitch or Monnify in Nigeria. Government and donor contracts are invoice and bank transfer only |
| Joining other systems        | Our own documented API first. FHIR UK Core read-only view in version 2. DHIS2 export for Nigeria. Hospital record interfaces where the vendor supports them                                                                      | Mostly effort: £20k–£40k per NHS integration, £30k–£50k for the FHIR view                                                  | Zapier or Make for business tools only, never for patient data                                                                                                |
| Cloud and databases          | AWS eu-west-2 with separate accounts for production, non-production, backups and security. PostgreSQL everywhere. Redis for queues only. S3 with write-once storage for audit copies                                             | See the product row. A Nigeria cell adds £350–£1,500                                                                       | Azure UK South. Hetzner or OVH are cheap but weak for NHS assurance. Vercel and Supabase are fine for the corporate site, not for patient data                |
| AI tools                     | Business tiers with no-training terms, for drafting marketing and course content, code help, internal meeting notes and translation first drafts                                                                                 | £20–£60 per user, plus £10–£40 for coding assistants                                                                       | None needed. What matters is the policy, not the vendor                                                                                                       |
| Security                     | Google Workspace sign-in with passkeys. Device management. 1Password. Email authentication set to reject forgeries. Cyber Essentials, then Plus, then the NHS toolkit                                                            | 1Password £6–£8 per user; devices £0–£8 each; certification £2k–£4k a year                                                 | Bitwarden instead of 1Password. ISO 27001 deferred to year 2 or 3 at £15k–£40k                                                                                |

| What it is for              | What to use                                                                                                                                                                                                                                                                                                                                                                                                    | Roughly per month                                                                                     | The alternative                                 |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Data governance and backups | ICO registration on incorporation, a fractional data protection officer, records of processing, retention schedule, DPIA templates and a sub-processor register. Product backups: point-in-time recovery, 35-day snapshots, a copy in a separate UK account, restore drills every quarter. Business tools export monthly to encrypted storage; code sits in GitHub with branch protection and a nightly mirror | DPO £500–£1,500; ICO fee £52–£78 a year; SaaS backup £3–£6 per user; product backup included in cloud | None. Neither is optional once real data exists |

Annual items on top (ESTIMATE): Cyber Essentials £300–£600; Cyber Essentials Plus £1,500–£3,000; penetration test £6,000–£12,000; accessibility audit £3,000–£6,000; ICO fee £52–£78; insurance £1,000–£3,000.

## The lean first-year setup

Phase A is months 1 to 3, services only, two paid seats. Phase B is months 4 to 12, with the MaternaLink build running, four or five seats.

| Component                                   | Phase A per month | Phase B per month             |
|---------------------------------------------|-------------------|-------------------------------|
| Domain, DNS, email security                 | £2                | £2                            |
| Google Workspace                            | £10–£22           | £25–£55                       |
| Website (Webflow)                           | £18–£25           | £18–£25                       |
| Website analytics (Plausible)               | £9                | £9                            |
| CRM (HubSpot)                               | £0–£20            | £20–£40                       |
| Finance (Xero, with Stripe and GoCardless)  | £15–£30           | £15–£30                       |
| Work management (Notion, GitHub Projects)   | £0–£16            | £16–£40                       |
| Contracts (PandaDoc or Docusign)            | £0–£25            | £25–£40                       |
| Password manager (1Password)                | £12–£16           | £30–£40                       |
| Scheduling (Calendly)                       | £0–£10            | £0–£10                        |
| AI assistants                               | £20–£60           | £80–£200                      |
| Course platform                             | £0                | £30–£120, from about month 6  |
| GitHub Team                                 | £0                | £12–£20                       |
| AWS, as environments come online            | £0                | £150–£700                     |
| Monitoring (Sentry, Grafana, uptime checks) | £0                | £20–£80                       |
| SMS gateways, sandbox then pilot            | £0                | £10–£150, billed on in pilots |
| <b>Tools total</b>                          | <b>£90–£250</b>   | <b>£460–£1,560</b>            |
| Fractional data protection officer          | £0–£500           | £500–£1,500                   |
| <b>Total including that officer</b>         | <b>£90–£750</b>   | <b>£960–£3,060</b>            |

## Do not build this yet

| Item                                        | Why not now                                                                            | Build it when                                                         |
|---------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| A custom website or CMS                     | No engineers. Webflow is enough                                                        | The engineering team exists and the brand has settled                 |
| A custom course platform                    | Bought platforms cover years 1 and 2                                                   | Learning revenue passes £150k a year, or a buyer demands self-hosting |
| A customer portal                           | There are no customers to serve                                                        | Five or more paying organisations                                     |
| A data warehouse                            | The volumes are tiny. Metabase and a spreadsheet are enough                            | More than five organisations, or research data needs it               |
| Native mobile apps                          | The web app meets the need. App stores add clinical safety re-testing on every release | Evaluation shows the web app cannot do it                             |
| NHS interfaces and FHIR                     | There is no NHS customer, and onboarding is heavy                                      | A trust asks for it, in version 2                                     |
| Hosting inside Nigeria                      | No contract requires it                                                                | A signed contract has a data residency clause                         |
| Two live regions at once                    | Cost and complexity                                                                    | A contract demands better than 99.9% availability                     |
| AI on patient data                          | No DPIA, no in-region endpoint, no clinical review process                             | Version 1.1 at the earliest, under the written policy                 |
| ISO 27001                                   | NHS buyers ask for Cyber Essentials Plus and the toolkit first                         | A buyer requires it, or year 3                                        |
| Kubernetes, or splitting into microservices | Managed containers and one well-organised application are faster and safer             | Many services with someone to run them, or a team past ten engineers  |
| A separate sign-in service for women        | Phone code plus PIN is enough                                                          | Merging it with staff sign-in is clearly worth it                     |
| Blockchain, tokens, own-brand wearables     | No user needs them and they damage credibility                                         | Never, unless evidence appears                                        |

## Rules that do not bend

1. Patient data lives only in the MaternaLink platform, in London, behind row-level security, audit and encryption. Business tools never receive it.
2. Every outside service is a sub-processor: it needs an agreement, a register entry and a DPIA update.
3. Every integration sits behind an interface, so a customer's choice of SMS provider or hospital system is configuration, not code.
4. Customer exports use documented formats, so nobody is locked in.
5. Monitoring, backups and infrastructure-as-code start with the first environment. They are the cheapest assurance evidence you will ever buy.

## Words explained

| Word | What it means                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| CMS  | The tool that lets a non-developer edit website pages. A headless CMS stores the content separately from the site that displays it |
| CRM  | The tool that tracks who you have spoken to and what you sold                                                                      |

| Word                      | What it means                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| Row-level security        | The database itself refuses to show one customer's rows to another                                                   |
| Sub-processor             | Any outside company that touches your data on your behalf                                                            |
| Data residency            | The country the data physically sits in                                                                              |
| Infrastructure-as-code    | The servers are described in files, so they can be rebuilt exactly                                                   |
| FHIR and DHIS2            | The standard NHS systems use to exchange health records, and the reporting system most African health ministries run |
| ICO                       | The UK data protection regulator. You register with it on incorporation                                              |
| ESTIMATE / TO<br>VALIDATE | Our best guess, not a fact / nobody has checked this yet                                                             |